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Develop a Simple Calculator using XML-RPC

AIM:

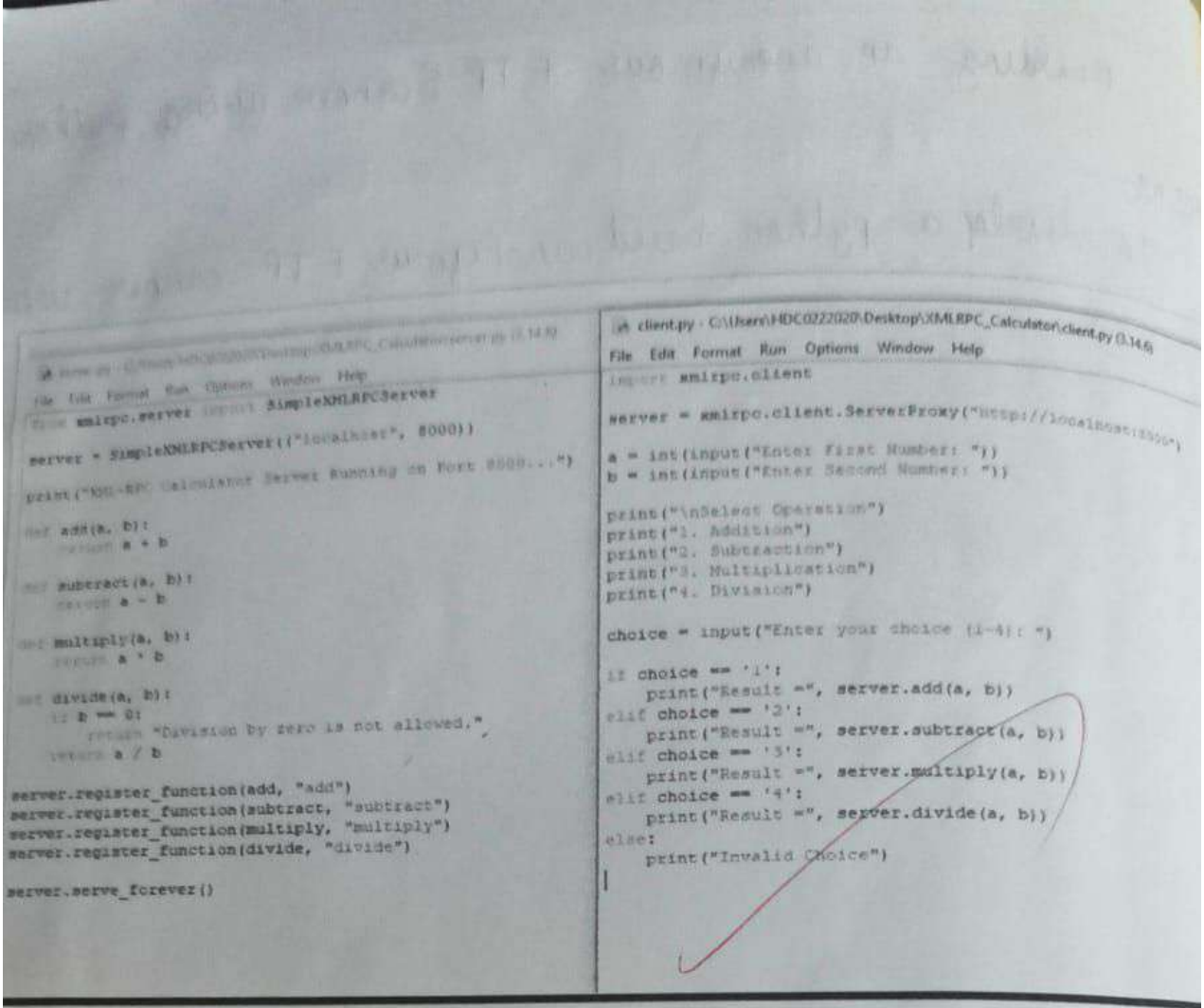
To develop a simple calculator using XML-RPC in python to perform basic arithmetic operations between a Client and a Server.

Procedure:

* Create a folder named XLR-RPC - calculator
* Create two files: server.py and client.py.
* Write the XML-RPC server.py.
* Write the server using python server.py
* Write the client program in client.py.
* Enter two numbers and select addition, Subtraction Multiplication & Division.

esult:

Thus, a simple calculator was successfully dev ing XML-RPC in python and arithmetic operati re performed between the client and server.



```
File Edit Format Run Options Window Help
from xmlrpc.server import SimpleXMLRPCServer

server = SimpleXMLRPCServer(("localhost", 8000))

print("XML-RPC Calculator Server Running on Port 8000...")

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed."
    return a / b

server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

server.serve_forever()
```

client.py - C:\Users\HDC0222020\Desktop\XMLRPC_Calculator\client.py (3.14.6)

```
File Edit Format Run Options Window Help
import xmlrpc.client

server = xmlrpc.client.ServerProxy("http://localhost:8000")

a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("\nSelect Operation")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = input("Enter your choice (1-4): ")

if choice == '1':
    print("Result =", server.add(a, b))
elif choice == '2':
    print("Result =", server.subtract(a, b))
elif choice == '3':
    print("Result =", server.multiply(a, b))
elif choice == '4':
    print("Result =", server.divide(a, b))
else:
    print("Invalid Choice")
```

C:\Windows\system32\cmd.exe - python server.py

```
:\Users\HDC0222020\Desktop\XMLRPC_Calculator>python server.py
ML-RPC Calculator Server Running on Port 8000...
27.0.0.1 - - [10/Sep/2026 08:31:32] "POST /RPC2 HTTP/1.1" 200 -
```

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C:\Windows\system32\cmd.exe

```
Users\HDC0222020\Desktop\XMLRPC_Calculator>

k to go back, hold to see history

op\XMLRPC_Calculator>python client.py
er First Number: 20
er Second Number: 10

ect Operation
Addition
Subtraction
Multiplication
Division
er your choice (1-4): 1
lt = 30

sers\HDC0222020\Desktop\XMLRPC_Calculator>
```